

Esther Laura Mukite Kituyi

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🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

Professional Summary

Electrical Engineer and Data Scientist specialising in predictive modelling, financial risk assessment, and smart-grid intelligence. Proven record of achieving top-tier leaderboard rankings on competitive data-science platforms (Zindi [Rank #1]). Experienced in building end-to-end ML pipelines, deploying AI solutions via APIs, and bridging hardware IoT systems with software machine learning. Passionate about applying Python, Deep Learning (CNNs), NLP, and Prompt Engineering to real-world business challenges.

Technical Skills

Languages & Frameworks	Python (NumPy, Pandas, Scikit-Learn, XGBoost), SQL (PostgreSQL), C++/Arduino
AI / ML Specialisations	NLP, Computer Vision (CNNs), Predictive Risk Modelling, Generative AI & Prompt Engineering
Data Engineering & MLOps	Medallion Architecture, BigQuery (GA4), APIs, Docker, Git, Streamlit
Visualisation & Tools	Power BI, Tableau, Matplotlib, Seaborn, Google Colab, VS Code, Excel (Expert)
Engineering	Power Systems, Smart Grid Technology, IoT Systems, MATLAB, Circuit Design

Experience

Ngao Labs Feb 2026 – Present
Data Science Trainee Nairobi, Kenya

- Develop and deploy end-to-end ML pipelines for financial risk assessment and expenditure prediction, using ensemble methods (XGBoost) and advanced feature engineering.
- Integrate AI models into applications via APIs, ensuring seamless data flow and performance monitoring.
- Collaborate cross-functionally to translate business requirements into production-ready AI solutions.

Women Techsters Fellowship (Tech4Dev) Sep 2024 – Present
Data Science & Engineering Fellow Remote

- Engineered a Smart Grid Anomaly Detection system in Python to identify technical faults and energy-theft signatures in unstructured sensor data.
- Implemented Medallion Architecture to streamline data pipelines from raw ingestion to executive-level Power BI dashboards.
- Analysed consumption patterns to optimise load balancing and improve grid reliability.

CCI Kenya (AT&T Campaign) Jul 2024 – Dec 2025
Technical Support Specialist Nairobi, Kenya

- Interpreted complex infrastructure logs to diagnose and resolve systemic technical issues for global clients, reducing average resolution time.

Centre for Transport and Road Engineering (CeTRE) Feb 2023 – May 2023
R&D Intern Nyeri, Kenya

- Assisted in data collection for e-mobility research to support green energy transition initiatives.

Selected Projects

Nairobi Traffic Prediction [Rank #1 — Ngao Labs] *Regression · Time-Series · APIs*

Developed a predictive pipeline integrating real-time weather and traffic API data. Applied Random Forest and time-series analysis to forecast urban congestion across Nairobi, achieving the highest accuracy score in the Ngao Labs capstone competition.

Tanzania Tourism Expenditure Prediction [Rank #1 — Zindi] *Regression · Ensemble*

Built an ensemble XGBoost model to predict visitor spending for the Tanzania Tourism Board. Engineered complex interaction features and custom “Luxury Segment” indicators to secure the top Zindi leaderboard position.

Loan Default Prediction — SuperLender Challenge [Top Rank — Zindi] *Classification · Credit Scoring*

Developed a credit scoring model to assess customer creditworthiness for a Kenyan fintech. Engineered novel “Interest Ratio” and “Loan History” features to predict default risk, achieving a top-tier leaderboard ranking on Zindi.

Smart Grid Anomaly Detection *Data Engineering · ML · ETL*

Engineered a Python-based system to detect technical faults and energy theft signatures in unstructured sensor data. Implemented Medallion Architecture (Bronze → Silver → Gold) to deliver clean, analytics-ready data for executive dashboards.

COVID-19 Vaccination Sentiment Analysis *NLP · Machine Learning*

Built an NLP pipeline to classify tweet sentiment on a continuous scale (−1 to +1). Engineered metadata features (hashtag and punctuation density) and integrated them with TF-IDF vectors using Ridge Regression, achieving high predictive accuracy.

Waste-to-Cash Image Validation *Computer Vision · Deep Learning*

Designed and trained a CNN (TensorFlow) to classify six categories of recyclable waste, enabling automated image validation for a “cash for trash” sustainability initiative. Augmented training data through synthetic dataset generation.

Air Quality Forecasting *Time-Series · Clustering · Forecasting*

Participated in a Zindi challenge to predict future air-pollution levels. Combined geospatial clustering with time-series forecasting models, rigorously evaluated on RMSE.

Financial Inclusion in Africa *Binary Classification · EDA*

Predicted bank account ownership across African demographic segments to improve financial accessibility. Applied deep demographic analysis and optimised binary classification models to identify key drivers of financial eligibility.

IoT Smart Metering System *Hardware · IoT*

Designed a GSM-enabled prototype for real-time utility monitoring, focusing on data collection and remote telemetry for energy optimisation (Arduino/C++ and GSM modules).

Education

Dedan Kimathi University of Technology (DeKUT)

Bachelor of Science in Electrical and Electronics Engineering

May 2019 – Apr 2024

Nyeri, Kenya